

Multi-material Topology Optimization of Permanent Magnet Motors Based on ON/OFF Method

Zhen Sun, *Student Member, IEEE*, Kota Watanabe, *Member, IEEE*

Graduate School of Engineering, Muroran Institute of Technology, Muroran, Japan, 20096506@mmm.muroran-it.ac.jp

This paper presents a multi-material topology optimization method based on a triangular cell-based ON/OFF method, filtering algorithm, and immune algorithm. Specifically, this method can be used to find out the optimal topology of electric machines that are composed of various materials, such as permanent magnet (PM), magnetic, and non-magnetic material. To validate the effectiveness of the proposed multi-material topology optimization method, the rotor of a benchmark PM synchronous motor is re-designed. The optimization results show that the torque property can be improved using as small amount of PM as possible, and the proposed optimization method is robust. The proposed multi-material topology method can be used in the conceptual design phase of PM motors. Moreover, this method can be easily integrated into commercial finite element analysis software that use Delaunay triangulation to discretize models.

Index Terms—Finite element analysis, immune algorithm, permanent magnet motors, topology optimization.

I. INTRODUCTION

TOPOLOGY optimization (TO) can find novel electric machines because it searches optimal material distribution by making free modifications in the distribution of material. For example, the rotor shape of interior permanent magnet (IPM) motors has been optimized with little dependence on engineers' knowledge by TO methods. The broad category of TO methods can be generally divided into the sensitivity-analysis-based method and the stochastic-algorithm-based method. The ON/OFF method is a typical stochastic-algorithm-based method [1]-[3]. In the ON/OFF method, the optimization model is subdivided into numerous small cells, the shapes are expressed by a binary coding, "ON (material)" or "OFF (no material)", of material in each cell, the optimal topology is obtained by trialing a greater variety of "ON"/"OFF" binary state combinations.

Although remarkable achievements have been made for ON/OFF-based TO, it is worth noting that most of the studies mentioned above are devoted to the problems with two materials. In modern engineering applications, however, the consideration of TO made in multi-material is unavoidable and has become increasingly concerned. Therefore, in this paper, we present a multi-material topology optimization method. This method uses a triangular cell-based ON/OFF method to treat up to three-material and adopt a specially designed filtering algorithm to ensure the optimized shapes are feasible for engineering realization. The immune algorithm (IA), which has strong exploration capability, is adopted to search the optimal material distribution. The proposed method can be used to optimize the distribution of three materials, such as magnet, iron, and air in electric machines.

This paper is organized as follows. In section II, the formulation of this multi-material ON/OFF method is introduced. In section III, this method is applied to the optimization of rotor shape in an IPM motor, in which the distributions of iron core, magnets, and flux barrier are optimized to improve the torque property as well as to save the magnet usage. In addition, the characteristic of the present method is discussed. Section IV gives the conclusions.

II. FORMULATIONS

A. Multi-material ON/OFF method

In the traditional ON/OFF method, the design region is discrete into rectangular cells, each cell can take up either the "ON" (solid) or "OFF" (void) state. As shown in Fig.1(a), the ON/OFF method often generates checkerboard structures, which have jagged boundaries and isolated material blocks. Thus, a post-process named filtering is commonly used to ensure the optimized topology is feasible for engineering realization by checking the material connection. As shown in Fig.1(b), however, the filtering process in the traditional rectangular cell-based ON/OFF method is difficult to tackle the checkerboard structure when treating more than two states.

In the proposed three-state ON/OFF method, the design region is discrete into triangular cells. As shown in Fig.2 (a), each cell can freely take up either "State1" (material1), "State2" (material2), or "State3" (material3). Similar to the traditional ON/OFF method, stochastic algorithms can be used for searching the optimal material distribution. Thanks to the triangular cells, three states in the checkerboard structure can be easily handled with the aid of the proposed neighborhood information-based filtering algorithm.

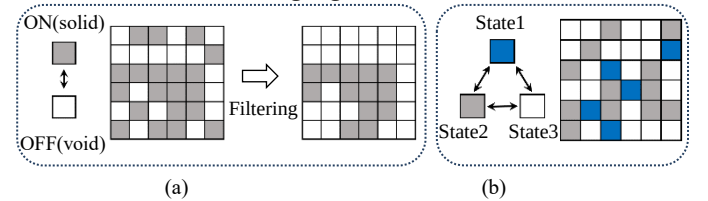


Fig. 1. Traditional rectangular cell-based binary-state ON/OFF method (a) Checkerboard structure and after filtering when treat two states (b) Checkerboard structure consisted of three states.

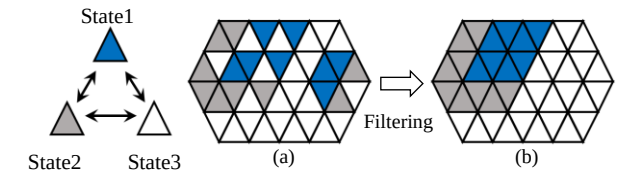


Fig. 2. Proposed triangular cell-based multi-material ON/OFF method (a) Checkerboard structure when treat three states (b) Checkerboard structure after filtering.

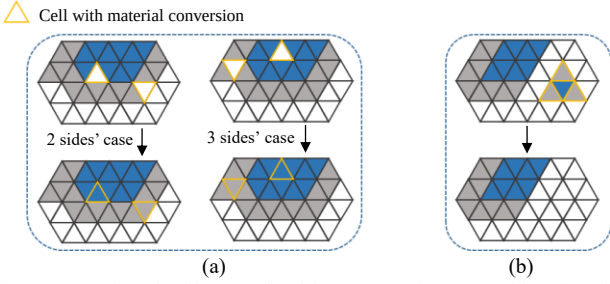


Fig. 3. Operations in filtering algorithm. (a) Surface smoothing (b) Floating material removing.

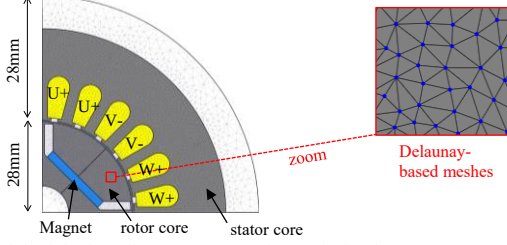


Fig. 4. Model of the baseline IPM motor for optimization. ($T_{ave}^{base}=2.15\text{Nm}$, $T_{rip}^{base}=1.03\text{Nm}$, $V_{base}=3.544\text{cm}^3$).

B. Filtering algorithm

The present method adopts a filtering algorithm to ensure the optimized shapes are feasible for realization. As shown in Fig.3, the filtering algorithm includes a surface smoothing process and a floating material removing process.

The surface smoothing process performs the following steps to ensure the smoothness of material's surface: (1) Search and check the neighborhood of each element. (2) If two or three sides of an element face toward one other state, the state of this element will be converted to fill the jagged boundary or cavity. (3) Back to Step 1 until no element can be converted.

The floating material removing process is performed after finishing the surface smoothing process, it has the following steps. (1) Select which material(s) should be managed (2) Check the material connection status of each cell, the connected cells with the same state (material) will be considered as a material block. (3) Calculate the area of each material block, list them up from big to small. (4) Keep the biggest material blocks, convert the other blocks to another producible material (usually air). (5) Back to Step 1 until no material(s) can be managed.

C. Immune algorithm

The IA is inspired by the clonal selection principle in the mammalian immune system. Compared with the genetic algorithm (GA), the IA has stronger exploitation capability realized by the affinity maturation. The detailed procedures of ON/OFF method with IA for TO can be found in [3].

III. OPTIMIZATION OF IPM MOTOR

A. Problem setting

A baseline IPM machine from IEEJ for testing numerical methods is chosen for optimization. Fig.3 shows the 1/4 model of this machine. Since the symmetry, $0^\circ - 45^\circ$ on the rotor, which includes 1263 elements, is the design region. To take the contribution to the torque from the reluctance torque into the rotor design, the machine is simulated by three-phase sinusoidal current whose amplitude is 3 Arms and phase angle

is 10° . During TO, magnetization direction of the PM block changes according to its shape. It is the vertical direction of the leftmost magnet element and the bottommost PM element.

B. Objective function

The distribution of magnets, magnetic core, and flux barrier in the design region is optimized. The Objective function is

$$F = \frac{T_{ave}}{T_{ave}^{base}} - \frac{T_{rip}}{T_{rip}^{base}} - \frac{V}{V_{base}} + p \quad (1)$$

$$\text{sub. to. } T_{ave} > T_{ave}^{base} \quad (2)$$

where T_{ave}^{base} , T_{rip}^{base} , and V_{base} are the average torque, torque ripple and PM volume of baseline machine, respectively. T_{ave} , T_{rip} , and s denote the average torque, torque ripple, and PM volume. p is a penalty function to restrict the average torque.

C. Optimization results

To clarify the effect of stochasticity on TO results, the IA was performed with three different initial random population groups (initial population A, B, C). The TO performed 300 generations in 80 hours. The resultant optimal topology A, topology B, topology C evolved by the three population groups are shown in Fig.5. It can be observed that the resultant shapes have some common features, which are consistent with the motor design intuition: (1) V-shaped magnets, which can enhance the reluctance torque, are obtained; (2) Slim magnets are formed; (3) Flux barriers (air regions) are emerged and formed in reasonable place to guide the flux.

As the result, all the optimal topologies have better performance than the baseline motor in terms of average torque, torque ripple, and PM consumption. In general, the torque ripple is decreased by 70% and the PM consumption is reduced by 40% compared with the baseline machine.

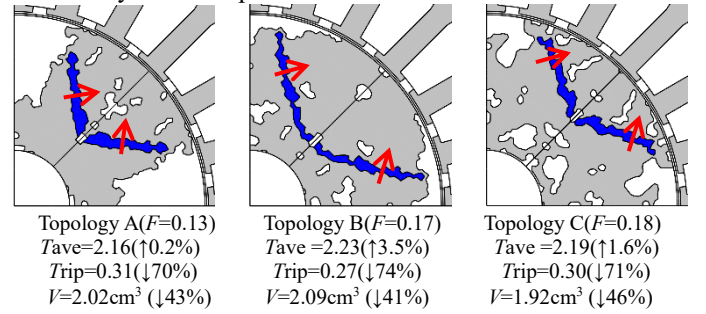


Fig. 5. Optimal rotor topologies obtained by the proposed method.

IV. CONCLUSIONS

We have presented a multi-material topology optimization method. The optimization results on an IPM motor show the effectiveness of this method since the improvement of torque property and decrease of magnet consumption. This method can provide guidance for the initial shape design of magnet and flux barrier in permanent magnet motors.

REFERENCES

- [1] N. Takahashi, T. Yamada, S. Shimose, and D. Miyagi, "Optimization of rotor of actual IPM motor using ON/OFF method," *IEEE Trans. Magn.*, vol. 47, no. 5, pp. 1262–1265, May 2011.
- [2] K. Watanabe et al., "Optimization of inductors using evolutionary algorithms and its experimental validation," *IEEE Trans. Magn.*, vol. 46, no. 8, pp. 3393–3396, Aug. 2010.
- [3] K. Watanabe, T. Suga and S. Kitabatake, "Topology Optimization Based on the ON/OFF Method for Synchronous Motor," *IEEE Trans. Magn.*, vol. 54, no. 3, pp. 1–4, Mar. 2018.